Machine Learning - Graduate Level Assessment 19 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. As of 2020, which architecture is best for classifying high-resolution images?
   (a) convolutional networks
   (b) graph networks
   (c) fully connected networks
   (d) RBF networks
2. MLE estimates are often undesirable because
   (a) they are biased
   (b) they have high variance
   (c) they are not consistent estimators
   (d) None of the above
3. K-fold cross-validation is
   (a) linear in K
   (b) quadratic in K
   (c) cubic in K
   (d) exponential in K
4. Which one of the following is the main reason for pruning a Decision Tree?
   (a) To save computing time during testing
   (b) To save space for storing the Decision Tree
   (c) To make the training set error smaller
   (d) To avoid overfitting the training set
5. Which of the following is NOT supervised learning?
   (a) PCA
   (b) Decision Tree
   (c) Linear Regression
   (d) Naive Bayesian

True / False (10 marks)

Answer True or False

6. True or False: A larger number of test examples is needed to achieve statistically significant results when the error rate is smaller.

7. True or False: Transforming the data to zero median is sufficient to obtain the same projection as Singular Value Decomposition (SVD) in PCA.

8. True or False: ImageNet contains images of various resolutions, making it a diverse dataset for training deep learning models.

9. True or False: The dimensionality of the null space of the matrix A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]] is 2.

10. True or False: Both Bayesians and frequentists agree on the necessity of using prior distributions in probabilistic regression models.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to select the nth items of a list.

12. Write a function to find numbers within a given range where every number is divisible by every digit it contains.

End of Paper